

Maxwell Macroscopic Equations.

First equation: is usually called the Gauss' Law expressing the relationship between the source density and displacement field

$$\nabla \cdot \vec{D} = \rho_{\text{free}}$$

In the integral form, we can write

$$\int_{\text{Volume}} \nabla \cdot \vec{D} \, dV = \int_{\text{Volume}} \rho_{\text{free}} \, dV = \int_{\text{Surface}} \vec{D} \cdot d\vec{S} = Q_{\text{free}}$$

The source of $\vec{D}$ field comes from source density

Second equation: is usually called the Ampere-Maxwell equation

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

The $\frac{\partial \vec{D}}{\partial t}$ is sometimes called the displacement current density with two parts.

- $\frac{\partial \vec{E}}{\partial t}$ is electric current density produced by local field $\vec{E}$.
- $\frac{\partial \vec{P}}{\partial t}$ is electric current density produced by bound electric charges.

Continuity equation of charge and current: the first and second equation give

$$\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{J} + \nabla \cdot \left[\frac{\partial \vec{D}}{\partial t} \right]$$

Left side is zero and the right hand side is first equation

$$\nabla \cdot \vec{J}_{\text{free}} + \frac{\partial \rho_{\text{free}}}{\partial t} = 0$$

The integral has the value

$$\int_{\text{Volume}} \nabla \cdot \vec{J}_{\text{free}} \, dV + \frac{d}{dt} \int_{\text{Volume}} \rho_{\text{free}} \, dV = 0$$

$$\int_S \vec{J}_{\text{free}} \cdot d\vec{S} = - \frac{d}{dt} Q_{\text{free}}$$

This equation indicates the interdependent nature of free charge and free current. Current must be produced by moving charges.

Third equation: is called the Faraday's law. It follows.

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\oint_{\text{loop}} \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_{\text{Surface}} \vec{B} \cdot d\vec{S}$$



This is the same as shaking the magnets to produce a time varying electric field.

Fourth equation: there is no evidence of magnetic monopoles.

$$\nabla \cdot \vec{B} = 0$$

$$\oint \vec{B} \cdot d\vec{S} = 0$$